

🔗 API Documentation: TaskFlow API

🔗 1. Overview

TaskFlow API is a task management web service developed for the `XJC03011 Web Services and Web Data` coursework. The application is built with **Django**, **Django REST Framework**, and **SQLite**. It provides full CRUD functionality for a single resource, `Task`, and returns JSON responses for API requests.

In addition to the JSON API, the project also includes simple HTML demonstration pages to make the application easier to navigate and present during the oral examination.

🔗 2. Deployment and Base URLs

🔗 Local Base URL

When running locally:

- `http://127.0.0.1:8000`

🔗 Live Deployment Base URL

The project is also deployed on PythonAnywhere:

- `https://yunshen.pythonanywhere.com`

🔗 Main Live Entry Pages

- Task home page: `https://yunshen.pythonanywhere.com/api/tasks/`
- Task selector page: `https://yunshen.pythonanywhere.com/api/tasks/select/`
- Raw API list endpoint: `https://yunshen.pythonanywhere.com/api/tasks/api/`

🔗 3. Technologies Used

- Python
- Django
- Django REST Framework
- SQLite
- Django ORM
- Django templates for demonstration pages
- PythonAnywhere for deployment

🔗 4. Data Model

🔗 Task

Field	Type	Required	Description
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id	integer	No	Unique task identifier
title	string	Yes	Task title
description	string	No	Optional task description
status	string	Yes	Task status: pending , in_progress , completed
priority	string	Yes	Priority: low , medium , high
due_date	datetime	No	Optional due date
created_at	datetime	No	Timestamp when task was created
updated_at	datetime	No	Timestamp when task was last updated

🔗5. JSON API Endpoints

The main API follows a REST-style structure for the `Task` resource.

🔗5.1 Get All Tasks

Endpoint: `GET /api/tasks/api/`

🔗Local URL

- `http://127.0.0.1:8000/api/tasks/api/`

🔗Live URL

- `https://yunshen.pythonanywhere.com/api/tasks/api/`

Returns all tasks stored in the database.

🔗Example response

```
[
  {
    "id": 1,
    "title": "Finish coursework report",
    "description": "Write the final draft",
    "status": "pending",
    "priority": "high",
    "due_date": "2026-04-25T12:00:00Z",
    "created_at": "2026-04-20T10:00:00Z",
    "updated_at": "2026-04-20T10:00:00Z"
  }
]
```

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Status codes

- `200 OK`

5.2 Create a Task

Endpoint: `POST /api/tasks/api/`

Local URL

- `http://127.0.0.1:8000/api/tasks/api/`

Live URL

- `https://yunshen.pythonanywhere.com/api/tasks/api/`

Creates a new task.

Example request body

```
{
  "title": "Finish coursework report",
  "description": "Write the final draft",
  "status": "pending",
  "priority": "high",
  "due_date": "2026-04-25T12:00:00Z"
}
```



Example success response

```
{
  "id": 1,
  "title": "Finish coursework report",
  "description": "Write the final draft",
  "status": "pending",
  "priority": "high",
  "due_date": "2026-04-25T12:00:00Z",
  "created_at": "2026-04-20T10:00:00Z",
  "updated_at": "2026-04-20T10:00:00Z"
}
```



Status codes

- `201 Created`
- `400 Bad Request`

🔗 5.3 Get One Task

Endpoint: `GET /api/tasks/<id>/`

🔗 Example pattern

- `GET /api/tasks/1/`

🔗 Local example

- `http://127.0.0.1:8000/api/tasks/1/`

🔗 Live example

- `https://yunshen.pythonanywhere.com/api/tasks/1/`

Returns one task by ID.

🔗 Path parameter

- `<id>` : integer task ID

🔗 Example success response

```
{
  "id": 1,
  "title": "Finish coursework report",
  "description": "Write the final draft",
  "status": "pending",
  "priority": "high",
  "due_date": "2026-04-25T12:00:00Z",
  "created_at": "2026-04-20T10:00:00Z",
  "updated_at": "2026-04-20T10:00:00Z"
}
```



🔗 Status codes

- `200 OK`
- `404 Not Found`

🔗 5.4 Update a Task

Endpoint: `PUT /api/tasks/<id>/`

🔗 Example pattern

- `PUT /api/tasks/1/`

🔗 Local example

- `http://127.0.0.1:8000/api/tasks/1/`

Live example

- `https://yunshen.pythonanywhere.com/api/tasks/1/`

Updates an existing task.

Example request body

```
{
  "title": "Finish coursework report",
  "description": "Submit the final version",
  "status": "completed",
  "priority": "medium",
  "due_date": "2026-04-25T12:00:00Z"
```



Example success response

```
{
  "id": 1,
  "title": "Finish coursework report",
  "description": "Submit the final version",
  "status": "completed",
  "priority": "medium",
  "due_date": "2026-04-25T12:00:00Z",
  "created_at": "2026-04-20T10:00:00Z",
  "updated_at": "2026-04-20T11:00:00Z"
```



Status codes

- `200 OK`
- `400 Bad Request`
- `404 Not Found`

5.5 Delete a Task

Endpoint: `DELETE /api/tasks/<id>/`

Example pattern

- `DELETE /api/tasks/1/`

Local example

- `http://127.0.0.1:8000/api/tasks/1/`

Live example

- `https://yunshen.pythonanywhere.com/api/tasks/1/`

Deletes a task by ID.

Success response

No response body.

Status codes

- 204 No Content
- 404 Not Found

6. HTML Demonstration Pages

These pages are not the primary JSON API endpoints, but they are included to make the project easier to use and demonstrate during the oral exam.

6.1 Task Home

Endpoint: `GET /api/tasks/`

Live page

- `https://yunshen.pythonanywhere.com/api/tasks/`

Purpose

- displays existing tasks
- includes a form to create a new task
- includes navigation buttons to other pages

6.2 Task Selector

Endpoint: `GET /api/tasks/select/`

Live page

- `https://yunshen.pythonanywhere.com/api/tasks/select/`

Purpose

- displays clickable task buttons for navigation
- provides a simpler view for selecting task detail pages

6.3 Task Detail Page

Endpoint: `GET /api/tasks/page/<id>/`

🔗 Example live page

- `https://yunshen.pythonanywhere.com/api/tasks/page/1/`

🔗 Purpose

- displays one task in a user-friendly format
- includes navigation to other tasks and pages
- supports smoother oral demonstration than raw JSON alone

🔗 7. Validation Rules

The API currently applies the following validation rules:

- `title` is required and cannot be blank
- `status` must be one of:
 - `pending`
 - `in_progress`
 - `completed`
- `priority` must be one of:
 - `low`
 - `medium`
 - `high`
- `due_date` is optional

🔗 8. Error Handling

The API uses standard HTTP status codes to communicate errors.

Status Code	Meaning
200	Request completed successfully
201	Resource created successfully
204	Resource deleted successfully
400	Invalid request data
404	Requested task not found

🔗 9. Authentication

No authentication is implemented in the current version of the project.

This is acceptable for the current coursework prototype, but authentication would be an important future improvement if the API were extended for real multi-user use.

🔗 10. Manual Testing Summary

The API has been manually tested in local development and checked again after deployment.

Manual testing included:

- task creation
- task retrieval (all tasks)
- task retrieval (single task)
- task update
- task deletion
- invalid input handling
- missing task ID handling
- page navigation between the custom HTML pages
- checking that the live PythonAnywhere deployment loads successfully
- checking that hosted routes match the documented endpoints

🔗11. Deployment Note

The live deployment on PythonAnywhere is intended for coursework presentation and demonstration. The hosted version provides an accessible online instance of the project, while the local version remains the main development environment.